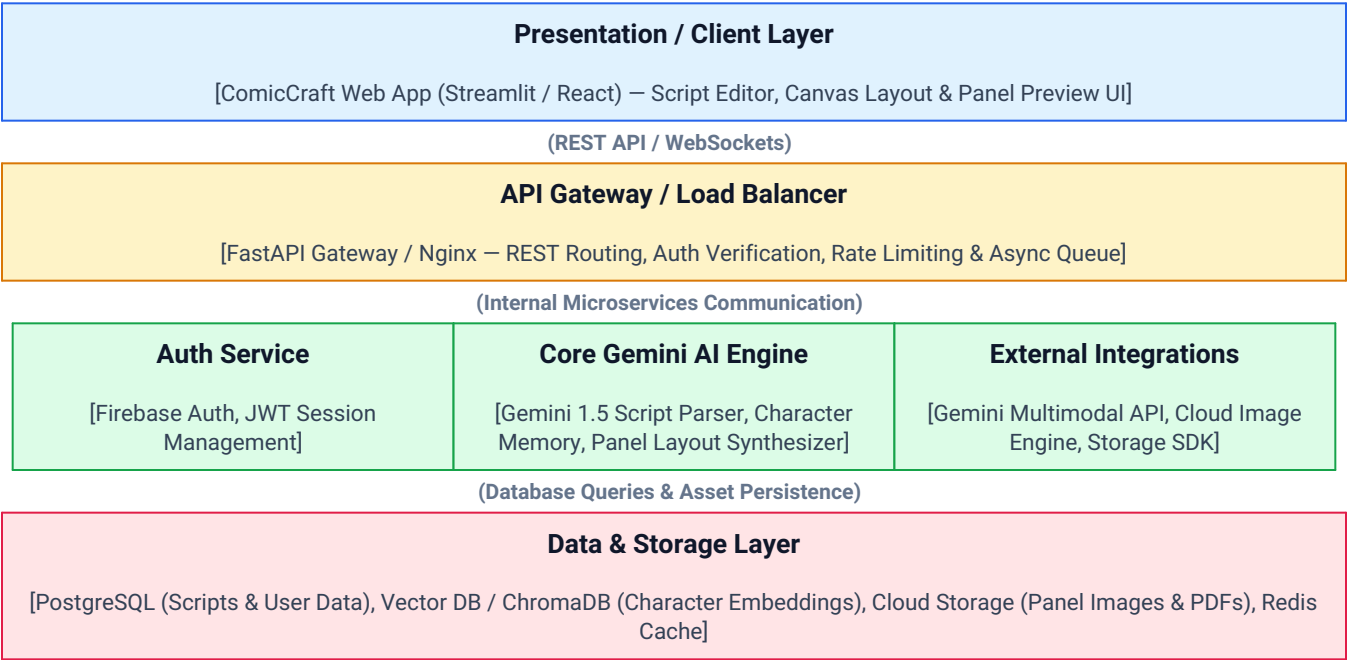


Solution Architecture Diagram & Specification

Architectural layout and component specifications for ComicCraft Gemini models

Project Name:	ComicCraft - AI Comic Story Creator using Gemini Models	Date:	30 September 2026
Team ID:	SWTID-2026-5650	Maximum Marks:	5 Marks

1. High-Level Architectural Diagram



2. Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Provides an interactive web user interface for story input, script editing, character style configuration, panel layout customization, and real-time comic reading.	Streamlit / React, Tailwind CSS, HTML5 Canvas
API Gateway	Manages REST API request routing, authentication verification, rate limiting, and asynchronous payload orchestration between client UI and core AI services.	FastAPI, Nginx, Uvicorn
Auth Service	Handles user authentication, session security, subscription tier checks, and project access permissions.	Firebase Auth, JWT, OAuth 2.0
Core Gemini AI Engine	Main processing engine that uses Gemini 1.5 Pro to parse raw scripts into structured panel JSON, enforces sequential character consistency, generates image prompts, and adds speech bubbles.	Python, Google Gemini 1.5 API, LangChain, Pillow, OpenCV

External API Integrations	Interfaces with Google Cloud Gemini multimodal endpoints and cloud asset pipelines for scalable image rendering and media delivery.	google-generativeai SDK, Google Cloud Storage API
Data & Storage Layer	Persists story metadata, user profiles, character visual embeddings, prompt logs, generated panel image assets, and compiled comic PDF exports.	PostgreSQL, ChromaDB / Vector DB, Redis Cache, Google Cloud Storage